

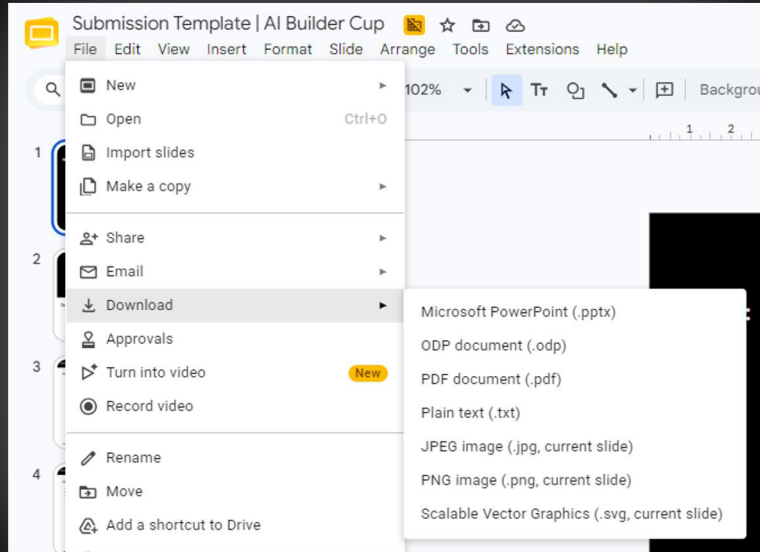


GOOGLE CLOUD AI BUILDER CUP 2026 · SUSTAINABILITY & SOCIAL IMPACT

Warning the junction constable before the siren arrives.

Team Details

- a. Team name: **Green Corridor**
- b. Team leader name: **Nandish**
- c. Problem Statement: **Sustainability & Social Impact — emergency vehicles stuck at red signals; the junction constable is never told**



Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

BRIEF ABOUT THE IDEA

Warn the junction constable early, sized to the queue

16.5 min

saved on the scripted
scenario

4 min

warning at a 400 m queue

100 %

tier accuracy on 11 synthetic
clips

Scripted scenario and synthetic voices, not field results.

OPPORTUNITIES

Not a fixed radius, and not waiting for new hardware



How it differs

Warning sized to the queue, not a fixed radius



How it solves

The constable clears the queue before the siren

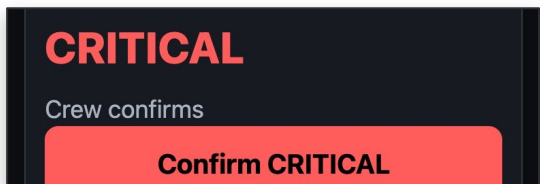


USP

Rules decide, Gemini explains; no hardware needed

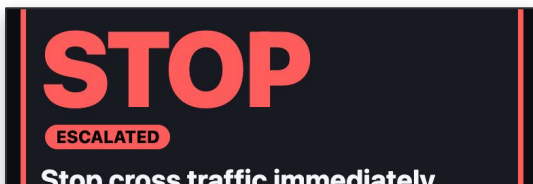
LIST OF FEATURES OFFERED BY THE SOLUTION

Six screens and the engines behind them



Voice triage

Speak, confirm the tier



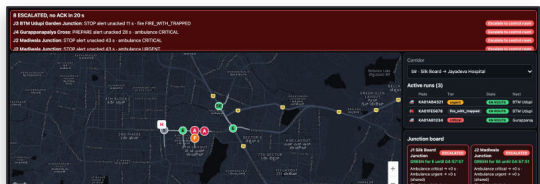
Spoken cop alert

PREPARE, then STOP



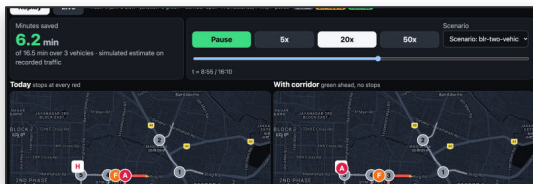
Hospital live brief

Live log, ATMIST



Control room board

Junctions and escalations



With-vs-without replay

Minutes saved

- Jayadeva Institute is the most suitable facility as it has both the required cath lab and ICU capabilities, with 4 beds available.
- It was chosen over Apollo Hospital because Apollo is currently on diversion, and Fortis does not have a cath lab.

- called required_capabilities(critical) -> cath_lab
- called list_hospitals(blir) -> 3 hospitals: blr_jayadeva, blr_apollo_bg, blr_fortis_bg
- called check_diversion(Jayadeva Institute of Cardiovascular Sciences) -> accepting
- called check_diversion(Apollo Hospital Bannerghatta Road) -> on diversion

Routing agent trace

Tool-call trace

Lead time

Sequencing

Cop voice

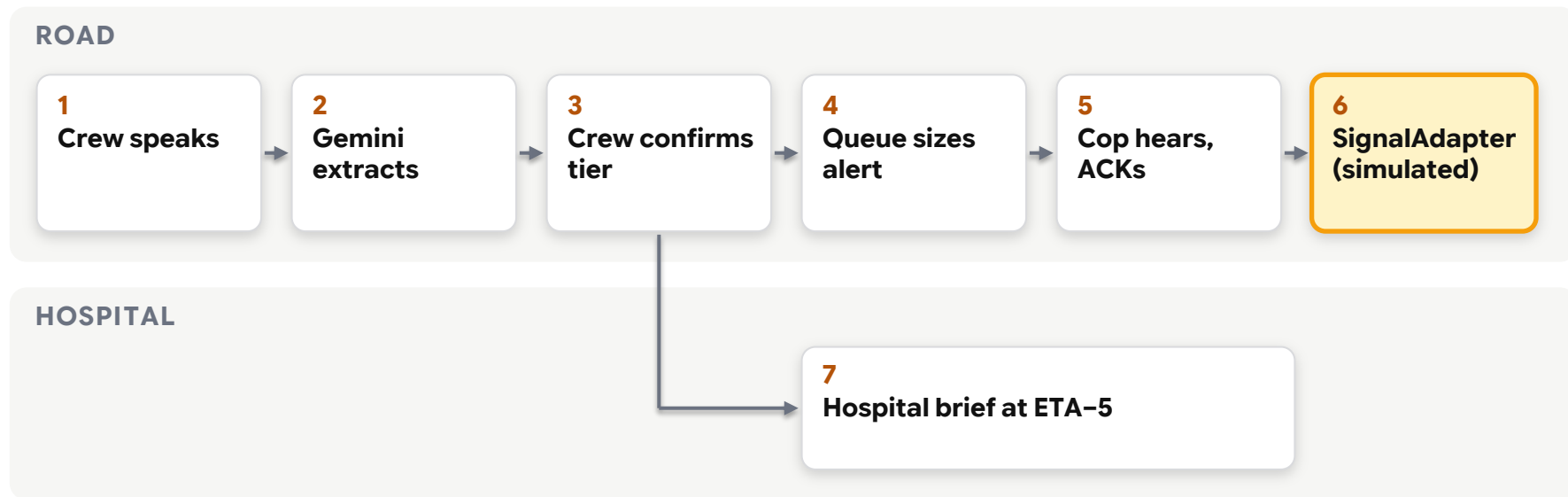
ATMIST

After-action

Eval harness

PROCESS FLOW DIAGRAM OR USE-CASE DIAGRAM

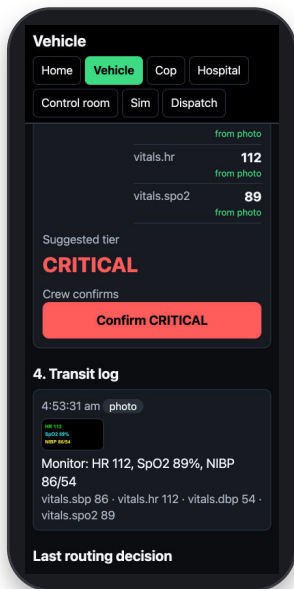
From the crew's words to the constable's phone



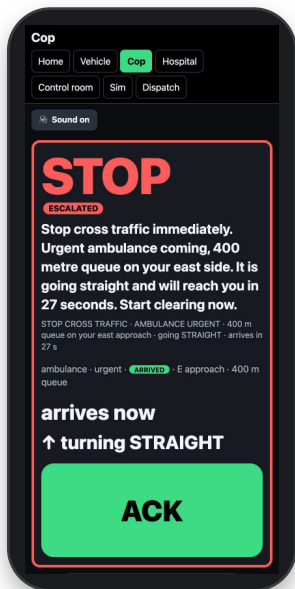
Amber seam: simulated today.

WIREFRAMES/MOCK DIAGRAMS OF THE PROPOSED SOLUTION (OPTIONAL)

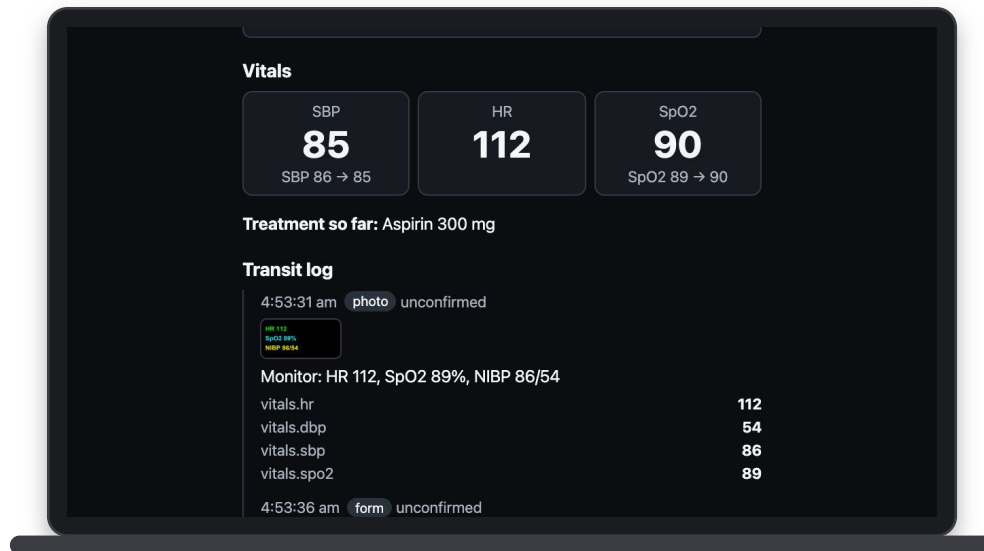
Real screens from the running prototype



Voice triage



Spoken alert

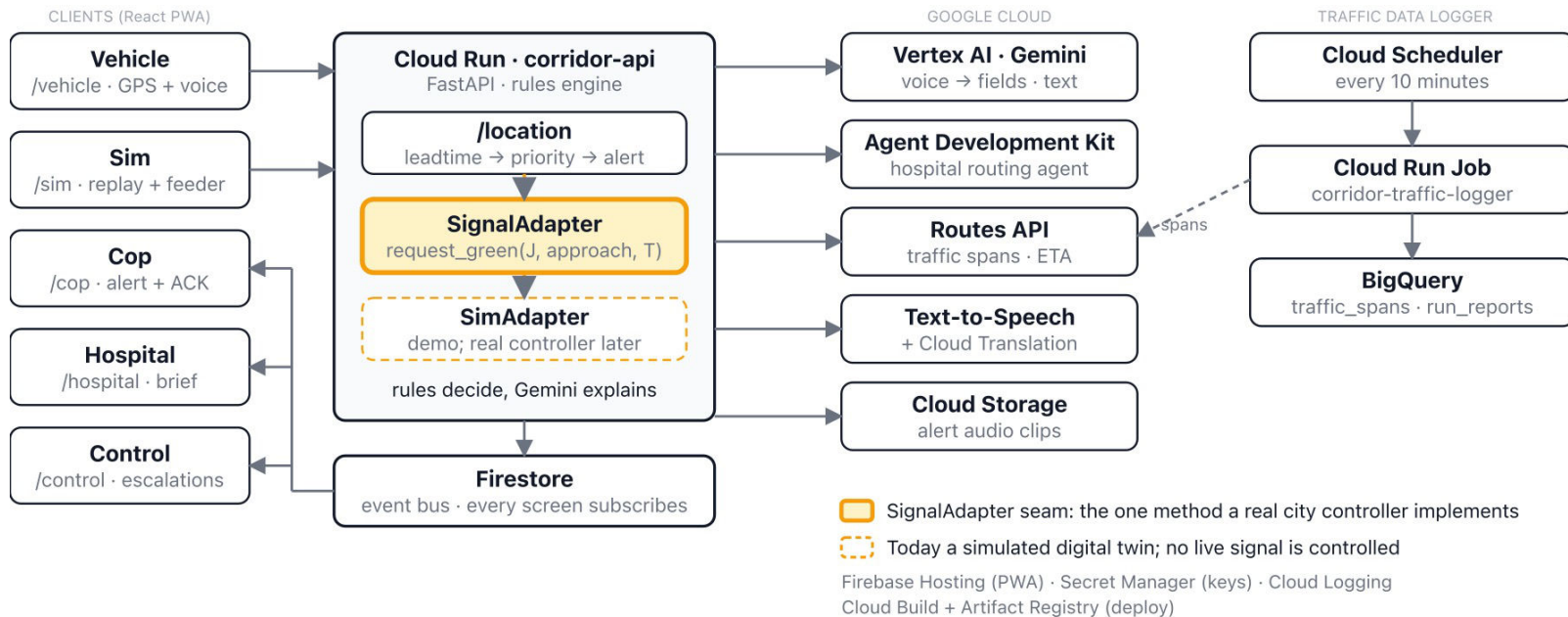


Hospital brief

Live rehearsal captures; synthetic patients.

ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION

Rules decide; Google Cloud services extract and explain



Rules engine on Cloud Run, Firestore event bus; the amber SignalAdapter seam is where a real controller plugs in.

TECHNOLOGIES TO BE USED IN THE SOLUTION

A Google Cloud stack, from model to delivery

● AI	Gemini on Vertex AI	Google ADK	BigQuery ML
● Run & data	Cloud Run	Firestore	BigQuery
● Maps & speech	Routes API	Text-to-Speech	Translation
● Delivery	Firebase Hosting	Cloud Build	Secret Manager

Also: Storage, Scheduler, Logging

ESTIMATED IMPLEMENTATION COST (OPTIONAL)

About ₹155 per vehicle-hour, almost all of it Routes

Service	Basis	₹ per hour
Routes	3 calls/min	₹153
Gemini	17 calls	₹0.34
Text-to-Speech	~1,000 chars	₹1.30

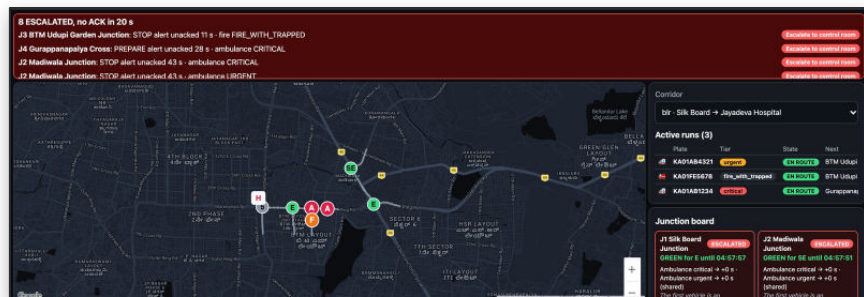
List prices; Routes is ~98 % of the bill.

≈ ₹155

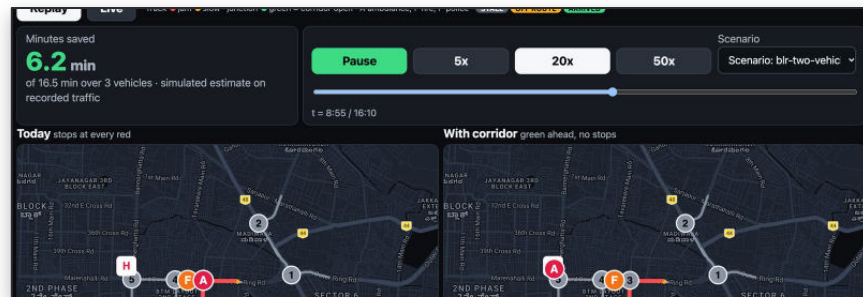
per vehicle-hour (estimate)

SNAPSHOTS OF THE PROTOTYPE

The loop, mid-run, from the live rehearsal



Control room



Replay simulator

- Jayadeva Institute is the most suitable facility as it has both the required cath lab and ICU capabilities, with 4 beds available.
 - It was chosen over Apollo Hospital because Apollo is currently on diversion, and Fortis does not have a cath lab.
- called required_capabilities(critical) → cath_lab
 - called list_hospitals(blr) → 3 hospitals: blr_jayadeva, blr_apollo_bg, blr_fortis_bg
 - called check_diversion(Jayadeva Institute of Cardiovascular Sciences) → accepting
 - called check_diversion(Apollo Hospital Bannerghatta Road) → on diversion

Routing trace

Vitals

SBP
85

SBP 86 → 85

HR
112SpO2
90

SpO2 89 → 90

Treatment so far: Aspirin 300 mg

Transit log

Hospital log

PROTOTYPE PERFORMANCE REPORT/BENCHMARKING

Synthetic-voice results, scripted-run timings

90%

field accuracy

100%

tier accuracy

3.5_s

mean extraction

4 vs 1min alert lead, long
vs short queue

11 synthetic clips; not field recordings.

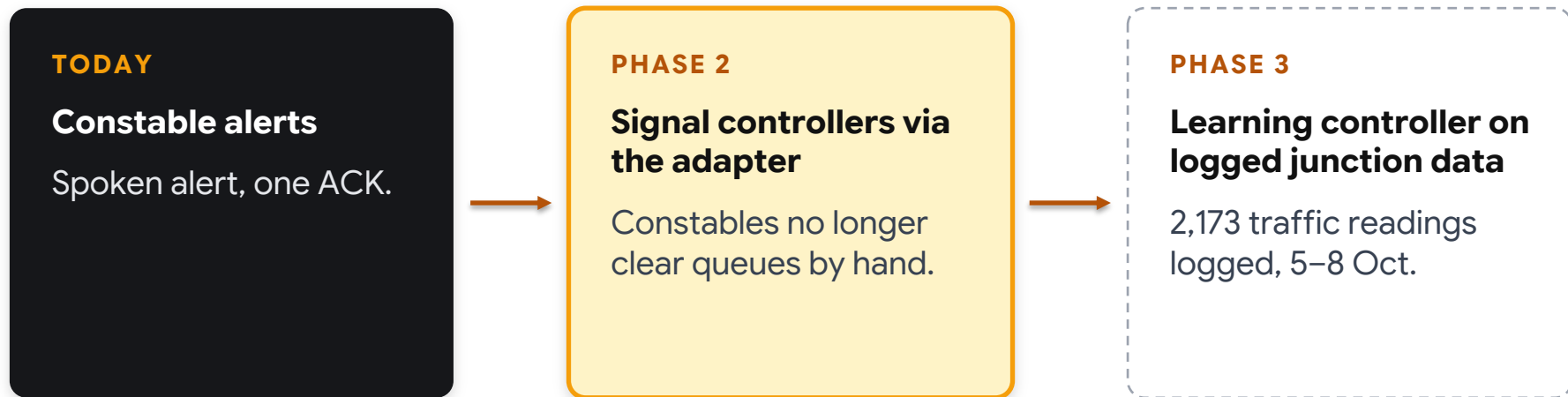
17 / 37_{ms}

/location p50 / p95, offline single instance

412 Python, 161 web tests; 98 % API coverage

ADDITIONAL DETAILS/FUTURE DEVELOPMENT (IF ANY)

From constable to controller, in three steps



PROVIDE LINKS TO YOUR:

Try it, read the code, watch the demo

GITHUB PUBLIC REPOSITORY



github.com/Nandish3010/ideal-disco

DEMO VIDEO (3 MINUTES)

[link]

FINAL PRODUCT



green-corridor-2026.web.app

Google Cloud ~~1125~~

AI  Builder
Cup Build in JAPAC.
Build for the world.

Thank you

